

GIPZIDE TABLET 5MG

CONTENT

Each tablet contains Glipizide 5 mg.

DESCRIPTION

White round tablet with a bisect and embossed with “SPS”, “G06” on one side, plain on the other.

PHARMACODYNAMICS

Glipizide is an oral blood-glucose lowering agent of the second-generation sulphonylureas group. When administered orally, glipizide stimulates insulin secretion from the beta cells of pancreatic islet tissue. As such, its action is dependent on the functioning of the beta cells in pancreatic islet tissue, which produce endogenous insulin. Stimulation of insulin secretion by glipizide is most pronounced in response to a meal. Long-term administration of glipizide does not affect the fasting insulin level, however, it enhances the postprandial insulin response following 6 months of treatment. The insulinotropic response to a meal occurs within 30 minutes after an oral dose of **Glipizide** in diabetic patients, but elevated insulin levels do not persist beyond the time of the meal challenge. Blood sugar control persists in some patients for up to 24 hours after a single dose of **Glipizide**, even though plasma levels have declined to a small fraction of peak levels by that time.

PHARMACOKINETICS

Glipizide is rapidly and completely absorbed after oral administration. Peak plasma concentration occurs 1 – 3 hours after a single oral dose. The elimination half-life for both oral administration and intravenous administration ranges from 2 – 4 hours in healthy subjects. First-pass metabolism effect for glipizide is not significant. When administered with meals, the absorption of glipizide is delayed by about 40 minutes, however, the total absorption and disposition of an oral dose is not affected. Glipizide is 98 – 99% bound to plasma protein, one hour after oral or intravenous administration. Metabolism of glipizide occurs mainly in the liver. The primary metabolites which are the inactive hydroxylation products and the polar conjugates are excreted mainly in the urine and less than 10% of glipizide is excreted unchanged.

INDICATIONS

Glipizide is indicated as an adjunct to diet and exercise to improve glycaemic control in adults with type 2 diabetes mellitus.

RECOMMENDED DOSE

As for any hypoglycaemic agent, dosage must be adapted for each individual case. Short-term administration of glipizide may be sufficient during periods of transient loss of control in patients usually controlled well on diet.

In general, glipizide should be given approximately 30 minutes before a meal to achieve the greatest reduction in postprandial hyperglycaemia.

Initial Dose

The recommended starting dose is 5 mg/day, given before breakfast or the mid-day meal. Elderly patients and other patients at risk for hypoglycaemia may be started on 2.5 mg.

Titration

Dosage adjustments should ordinarily be in increments of 2.5 mg or 5 mg, as determined by blood glucose response. At least several days should elapse between titration steps.

Maintenance

Some patients may be effectively controlled on a once-a-day regimen. The maximum recommended single dose is 15 mg. If this is not sufficient, splitting the daily dosage may prove effective. Doses above 15 mg should ordinarily be divided. Total daily dosage above 15 mg should ordinarily be divided. Total dosage above 30 mg has been safely given on a twice-a-day basis to long-term patients. Patients can usually be stabilized on a dosage ranging from 2.5 mg to 30 mg daily. The maximum recommended daily dosage is 40 mg.

Use in Children

Safety and effectiveness in children have not been established.

Use in Elderly and High-Risk Patients

To decrease the risk of hypoglycaemia in patients at risk including elderly, debilitated, and malnourished patients or patients with irregular caloric intake and patients with impaired renal or hepatic function, the initial and maintenance dosing should be conservative to avoid hypoglycaemic reactions.

Patients Receiving Insulin

As with other sulphonylurea-class hypoglycaemic, many stable type 2 diabetic patients receiving insulin may be transferred safely to treatment with glipizide.

When transferring patients from insulin to glipizide, the following general guidelines should be considered:

For patients whose daily insulin requirement is 20 units or less, insulin may be discontinued and glipizide therapy may begin at usual dosages. Several days should elapse between titration steps.

For patients whose daily insulin requirement is greater than 20 units, the insulin dose should be reduced by 50% and glipizide therapy may begin at usual dosages. Subsequent reductions in insulin dosage should depend on individual patient response. Several days should elapse between titration steps.

During the insulin withdrawal period, the patient should self-monitor glucose levels. Patients should be instructed to contact the prescriber immediately if these tests are abnormal. In some cases, especially when the patient has been receiving greater than 40 units of insulin daily, it may be advisable to consider hospitalization during the transition period.

Patients Receiving Other Oral Hypoglycaemic Agents

As with other sulphonylurea-class hypoglycaemics, no transition period is necessary when transferring patients to glipizide. Patients should be observed carefully (1 – 2 weeks) for hypoglycaemia when being transferred from longer half-life sulphonylureas (e.g., chlorpropamide) to glipizide due to potential overlapping of drug effect.

Combination Use

When adding other blood-glucose-lowering agents to glipizide for combination therapy, the agent should be initiated at the lowest recommended dose, and patients should be observed carefully for hypoglycaemia. Refer to the product information supplied with the oral agent for additional information.

When adding glipizide to other blood-glucose-lowering agents, glipizide can be initiated at 5 mg. Those patients who may be more sensitive to hypoglycaemic drugs may be started at a lower dose. Titration should be based on clinical judgment.

Method of administration: For oral use only.

CONTRAINDICATIONS

- Hypersensitivity to glipizide, other sulphonylureas or sulphonamides, or to any of the excipients in the formulation.
- Insulin-dependent diabetes mellitus, diabetic ketoacidosis, diabetic coma.
- Severe renal or hepatic insufficiency.
- Patients treated with miconazole.
- Pregnancy and lactation.

WARNINGS AND PRECAUTIONS

Glucose-6-phosphate dehydrogenase deficiency

Since glipizide belongs to the class of sulphonylurea agents, caution should be used in patients with G6PD deficiency. Treatment of patients with G6PD deficiency with sulphonylurea agents can lead to haemolytic anaemia and a non-sulphonylurea alternative should be considered.

Hypoglycaemia

All sulphonylurea agents are capable of producing severe hypoglycaemia. Renal or hepatic insufficiency may cause elevated blood levels of glipizide and the latter may also diminish gluconeogenic capacity, both of which increase the risk of serious hypoglycaemic reactions. Elderly, debilitated or malnourished patients and those with adrenal or pituitary insufficiency are particularly susceptible to the hypoglycaemic action of glucose-lowering drugs.

Hypoglycaemia may be difficult to recognise in the elderly, and in people who are taking beta-adrenergic blocking drugs. Hypoglycaemia is more likely to occur when caloric-intake is deficient, after severe or prolonged exercise, when alcohol is ingested, or when more than one glucose-lowering drug is used.

Loss of control of blood glucose

When a patient stabilised on a diabetic regimen is exposed to stress such as fever, trauma, infection, or surgery, a loss of control may occur. At such times, it may be necessary to discontinue glipizide and administer insulin.

The effectiveness of any oral hypoglycaemic drug, including glipizide, in lowering blood glucose to a desired level decreases in many patients over a period of time, which may be due to progression of the severity of diabetes or due to diminished responsiveness to the drug. This phenomenon is known as secondary failure, to distinguish it from primary failure in which the drug is ineffective in an individual patient when first given. Adequate adjustment of dose and adherence to diet should be assessed before classifying a patient as a secondary failure.

Renal and Hepatic Disease

The pharmacokinetics and/or pharmacodynamics of glipizide may be affected in patients with impaired renal or hepatic function. If hypoglycaemia should occur in such patients, it may be prolonged and appropriate management should be instituted.

Information for Patients

Patients should be informed of the potential risks and advantages of glipizide and of alternative modes of therapy. They should also be informed about the importance of adherence to dietary instructions, of a regular exercise program, and of regular testing of urine and/or blood glucose.

The risks of hypoglycaemia, its symptoms and treatment, and conditions that predispose to its development should be explained to patients and responsible family members. Primary and secondary failure should also be explained.

Laboratory Tests

Blood and urine glucose should be monitored periodically. Measurement of glycosylated haemoglobin may be useful.

Photosensitivity

Glipizide may cause the skin to be more sensitive to sunlight than it is normally. Although chances are rare, exposure to sunlight may cause skin rash, itching, redness or severe sunburn. Patients on **Gipzide** should protect their skin when exposed to the sun, especially between 10 am and 3 pm. If outdoors, wear protective clothing and use a minimum of SPF30+ sunscreen. Patients should be advised to inform their doctor if their skin does appear to be burning.

Effect on ability to drive and use machines

The effect of glipizide on the ability to drive or operate machines has not been studied; however, there is no evidence to suggest that glipizide may affect these abilities. Patients should be aware of the symptoms of hypoglycaemia and be careful about driving and the use of machines, especially when optimum stabilisation has not been achieved, for example during the change-over from other medications or during irregular use.

DRUG INTERACTIONS

The following products are likely to increase hypoglycaemic effect:

Contraindicated combinations

- **Miconazole**: Increase in hypoglycaemic effect, possibly leading to symptoms of hypoglycaemia or even coma.

Inadvisable combinations

- **Nonsteroidal anti-inflammatory drugs (e.g. phenylbutazone)**: Increase in hypoglycaemic effect of sulphonylureas (displacement of sulphonylurea binding to plasma proteins and/or decrease in sulphonylurea elimination).
- **Alcohol**: Increase in hypoglycaemic reaction, which can lead to hypoglycaemic coma.

Combinations requiring precaution

- **Fluconazole**: Increase in the half-life of the sulphonylurea, possibly giving rise to symptoms of hypoglycaemia.
- **Voriconazole**: Although not studied, voriconazole may increase the plasma levels of sulphonylureas, (e.g. tolbutamide, glipizide and glyburide) and therefore cause hypoglycaemia. Careful monitoring of blood glucose is recommended during co-administration.
- **Salicylates (acetylsalicylic acid)**: Increase in hypoglycaemic effect by high doses of acetylsalicylic acid (hypoglycaemic action of the acetylsalicylic acid).
- **Beta-blockers**: All beta-blockers mask some of the symptoms of hypoglycaemia (i.e. palpitations and tachycardia). Most non-cardioselective beta-blockers increase the incidence and severity of hypoglycaemia.
- **Angiotensin-converting enzyme inhibitors**: The use of angiotensin-converting enzyme inhibitors may lead to an increased hypoglycaemic effect in diabetic patients treated with sulphonylureas.
- **Cimetidine**: The use of cimetidine may be associated with a reduction in postprandial blood glucose in patients treated with glipizide.

The hypoglycaemic action of sulphonylureas, in general may also be potentiated by monoamine oxidase inhibitors, quinolones and drugs that are highly protein bound, such as sulphonamides, chloramphenicol, probenecid, coumarins and fibrates. When such drugs are administered to (or withdrawn from) a patient receiving glipizide, the patient should be observed closely for hypoglycaemia (or loss of control).

The following products could lead to hyperglycaemia:

Inadvisable combinations

- **Danazol**: Diabetogenic effect of danazol. If it cannot be avoided, warn the patient and step up self-monitoring of blood glucose and urine. Possibly adjust the dosage of anti-diabetic agent during treatment with danazol and after its discontinuation.

Combinations requiring precaution

- **Phenothiazines (e.g. chlorpromazine) at High Doses (> 100 mg/day of chlorpromazine)**: Elevation in blood glucose (reduction in insulin release).
- **Corticosteroids**: Elevation in blood glucose.
- **Sympathomimetics (e.g. ritodrine, salbutamol, terbutaline)**: Elevation in blood glucose due to beta-2-adrenoceptor stimulation.
- **Progestogens**: Diabetogenic effects of high-dose progestogens. Warn the patient and step up self-monitoring of blood glucose and urine. Possibly adjust the dosage of anti-diabetic agent during treatment with the neuroleptics, corticoids or progestogen and after discontinuation.

Other drugs that may produce hyperglycaemia and lead to a loss of control include the thiazides and other diuretics, thyroid products, oestrogens, oral contraceptives, phenytoin, nicotinic acid, calcium channel blocking drugs and isoniazid. When such drugs are administered to (or withdrawn from) a patient receiving glipizide, the patient should be observed closely for hypoglycaemia.

The risk of increase of effect of barbiturates is extremely low for pharmacokinetic reasons (non-ionic protein binding).

Tetracycline may interfere with determination of urine glucose. Cyclophosphamide and derivatives should also be used with care in diabetic patients since increased and decreased effects of sulphonylureas have been reported.

PREGNANCY AND LACTATION

Pregnancy

Glipizide is contraindicated in pregnancy.

Glipizide was found to be mildly foetotoxic in rat reproductive studies. No teratogenic effects were found in rat or rabbit studies. Prolonged severe hypoglycaemia (4 – 10 days) has been reported in neonates born to mothers who were receiving a sulphonylurea drug at the time of delivery.

Because recent information suggests that abnormal blood glucose levels during pregnancy are associated with a higher incidence of congenital abnormalities, many experts recommend that insulin be used during pregnancy to maintain blood glucose levels as close to normal as possible.

Lactation

No data are available on secretion into breast milk. Therefore glipizide is contraindicated in lactation.

SIDE EFFECTS

The majority of side effects have been dose related, transient, and have responded to dose reduction or withdrawal of the medication. However, clinical experience thus far has shown that, as with other sulphonylureas, some side effects associated with hypersensitivity may be severe and deaths have been reported in some instances. Aplastic anaemia and disulfiram-like reactions have been reported with other sulphonylureas.

The reported adverse reactions, which may possibly be associated with glipizide, are listed below:

Blood and lymphatic system disorders

Not known	Leukopenia, agranulocytosis, thrombocytopenia, haemolytic anaemia, pancytopenia
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Metabolism and nutrition disorders

Common	Hypoglycaemia
Not known	Hyponatremia

Psychiatric disorders

Not known	Confusional state #
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Nervous system disorders

Uncommon	Dizziness #, somnolence #, tremor #
Not known	Headache #

Eye disorders

Uncommon	Vision blurred #
Not known	Diplopia #, visual impairment #, visual acuity reduced #

Gastrointestinal disorders

Common	Nausea Δ, diarrhoea Δ, abdominal pain upper Δ, abdominal pain
Uncommon	Vomiting
Not known	Constipation Δ

Hepatobiliary disorders

Uncommon	Jaundice cholestatic %
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Not known Hepatic function abnormal, hepatitis

Skin and subcutaneous tissue disorders

Uncommon Eczema ^

Not known Dermatitis allergic ^, erythema ^, rash morbilliform ^, rash maculopapular ^, urticaria ^, pruritus ^, photosensitivity reaction

Congenital, familial and genetic disorders

Not known Porphyria non-acute

General disorders and administration site conditions

Not known Malaise #

Investigations

Not known Aspartate aminotransferase increased*, blood lactate dehydrogenase increased*, blood alkaline phosphatase increased*, blood urea increased*, blood creatinine increased*

This is usually transient and do not require discontinuance of therapy; however, they may also be symptoms of hypoglycaemia.

Δ Appear to be dose related and usually disappear on division or reduction of dosage.

% Discontinue treatment if cholestatic jaundice occurs.

^ They frequently disappear with continued therapy. However if they persist, the drug should be discontinued.

* The relationship of these abnormalities to glipizide is uncertain, and they have rarely been associated with clinical symptoms.

SYMPTOMS AND TREATMENT OF OVERDOSE

There is no well documented experience with glipizide overdosage. Overdosage of sulphonylureas including glipizide can produce glycaemia. Mild hypoglycaemic symptoms without loss of consciousness or neurologic findings should be treated actively with oral glucose and adjustments in drug dosage and/or meal patterns.

Close monitoring should continue until the physician is assured that the patient is out of danger. Severe hypoglycaemic reactions with coma, seizure, or other neurological impairment occur infrequently, but constitute medical emergencies requiring immediate hospitalisation. If hypoglycaemic coma is diagnosed or suspected, the patient should be given a rapid intravenous injection of concentrated (50 %) glucose solution. This should be followed by a continuous infusion of a more dilute (10 %) glucose solution at a rate that will maintain the blood glucose at a level above 100 mg/dL (5.55 mmol/L).

Patients should be closely monitored for a minimum of 48 hours and depending on the status of the patient at this time the physician should decide whether further monitoring is required. Clearance of glipizide from plasma may be prolonged in people with liver disease. Because of the extensive protein binding of glipizide, dialysis is unlikely to be of benefit.

STORAGE CONDITION

Store in a cool dry place, below 30°C.

Keep out of reach of children.

(Jauhi daripada kanak-kanak)

SHELF LIFE: Please refer to the outer box label.

PACKING/PACK SIZE

100 / 500 Tablets

MANUFACTURED BY

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